

Maincraft Technology

AIML Internship Task 2

Feature Engineering, Model Optimization & Performance Comparison

Submitted By: Sampada Bandre

Introduction

The objective of this project was to improve house price prediction performance and compare multiple machine learning regression models using the California Housing Dataset. The project focused on data preprocessing, feature scaling, model training, and performance evaluation.

Methodology

The California Housing Dataset was loaded from the Scikit-Learn library. The dataset contains information such as median income, house age, average rooms, average bedrooms, population, occupancy, latitude, longitude, and median house value.

Data exploration was performed using `head()`, `info()`, and `describe()` functions. Missing values were checked and no null values were found.

The target variable was `MedHouseVal`, while the remaining columns were used as input features.

Feature scaling was applied using `StandardScaler` to standardize the data and improve model performance. The dataset was then divided into training (80%) and testing (20%) sets using `train_test_split()`.

Three regression models were trained and compared:

- Linear Regression
- Ridge Regression
- Decision Tree Regressor

Results

The performance of each model was evaluated using:

- Root Mean Squared Error (RMSE)
- R^2 Score

Model Performance Comparison

Model	RMSE	R ² Score
Linear Regression	0.745581	0.575788
Ridge Regression	0.745554	0.575819
Decision Tree Regressor	0.724234	0.599732

A comparison chart and Actual vs Predicted scatter plot were generated to visualize model performance and prediction accuracy.

Based on the evaluation metrics, the best-performing model was selected and saved for future use.

Conclusion

This project successfully demonstrated feature engineering and model comparison techniques using the California Housing Dataset. Feature scaling improved data preparation, while multiple regression algorithms were evaluated to identify the most effective model.

The use of RMSE and R² Score provided a reliable method for comparing model performance. Overall, the project provided practical experience in data preprocessing, machine learning model training, evaluation, and model selection.

Future Scope

Future improvements may include:

- Hyperparameter tuning
- Cross-validation
- Random Forest Regression
- XGBoost Regression
- Model deployment using Flask or Streamlit

These techniques can further improve prediction accuracy and model performance.